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$$\begin{aligned} & \int \frac{dx}{x^2 - 4x + 13} \\ &= \int \frac{dx}{(x-2)^2 + 9} \\ &= \int \frac{dx}{9 \left(\frac{x-2}{3} \right)^2 + 1} \\ &= \frac{1}{9} \int \frac{dx}{\left(\frac{x-2}{3} \right)^2 + 1} \\ & t = \frac{x-2}{3} \\ & dx = 3dt \\ &= \frac{1}{3} \int \frac{dx}{t^2 + 1} \\ &= \frac{1}{3} \operatorname{Bgtan} \left(\frac{x-2}{3} \right) + C \end{aligned}$$

References